Can Diminishing Returns Restrain an AI Race?

Commercial Incentives, Delegation, and Resource Control

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October 21, 2026
Working paper

Abstract

I examine whether diminishing commercial returns to artificial intelligence can prevent dangerous autonomous development. A finite-horizon stochastic game links developers' research and delegation decisions to an autonomous system's subsequent resource choices. Governments regulate developers in two jurisdictions. Diminishing returns alone do not guarantee stopping: benefit levels, research costs, positional rewards, and continuation values determine whether developers delegate. Once control is lost, commercial curvature has no direct effect on a system whose objective and feasible resources are independent of revenue. I derive a global no-delegation certificate, an exact local delegation threshold, and conditions for acquisition to threaten human subsistence. A descriptive analysis of 27,471 Arena comparisons finds negative but imprecisely estimated curvature in preferences over measured capability; alternative matching and capability measures materially change that pattern. Cost and capability data discipline explicit calibration bridges, while control failure and resource conflict remain scenario assumptions. Quantitative results identify conditional regions of stopping and exposure rather than estimate humanity's extinction probability.

Keywords: artificial intelligence; endogenous stopping; technology races; delegation; existential risk.
JEL codes: C72, D62, D81, O31, O33.

Source code, frozen numeric data, and replication instructions:
https://github.com/dgonzalezgonzalez/pdoom-economics

1 Introduction

Can the economics of developing artificial intelligence provide a limit on its dangers? A commercial developer improves a model when the improvement raises product value, lowers costs, or strengthens its competitive position. If additional capability becomes less valuable, an optimally chosen development frontier may stop expanding. Whether that restraint operates before dangerous capabilities emerge is a separate question.

The distinction matters for recursive self-improvement: a process in which an AI system contributes to developing its successors. Recursive improvement need not produce an intelligence explosion, and an explosion is not necessary to create a dangerous system. The economic decisions include whether developers finance further research, whether they delegate decisions about that research, and whether an autonomous system must continue obtaining resources through markets that serve humans. A system pursuing another objective may value resources even when customers attach little additional value to its outputs.

I develop a model connecting these decisions. Two developers choose research, safety, and delegation under policies set by two governments. Delegated research can improve productivity and can also result in a loss of control. An uncontrolled system then chooses research and acquisition to maximize future capability. Diverting enough resources from human production can violate a subsistence constraint. Developers and governments anticipate this continuation problem. Catastrophe eliminates their future benefits, including commercial rents and the value of being alive.

The starting conjecture is that dangerous racing requires constant or increasing commercial returns. This is false without additional assumptions. A strictly concave benefit function can yield a privately optimal capability above a finite danger threshold. Conversely, sufficiently high research costs can prevent that threshold from being reached with constant or increasing benefits. In a dynamic race, the option value of better research technology and the value of overtaking a rival further separate commercial curvature from development incentives.

The constructive question is which restrictions make diminishing returns protective. I derive a benchmark threshold under a concave investment envelope and an explicit recursive comparison between commercial saturation and research-cost reduction. Even bounded benefits sustain perpetual improvement when recursive costs fall sufficiently quickly. A global certificate for a no-delegation equilibrium then evaluates deviations at every controlled state, including states reached after an earlier deviation. A local formula separates delegation's productivity gain from its expected loss of controlled continuation value. These conditions retain the original economic intuition while identifying what must be established in addition to concavity.

A second result concerns the boundary of that protection. Conditional on the state in which control is lost, commercial returns do not constrain a system if neither its preferences nor its feasible resources depend on those returns. Commercial curvature nevertheless changes whether, when, and with what capability control is lost. A financing constraint that remains enforceable after delegation can also restrict the system. The result identifies the assumption needed to move from weak customer demand to autonomous restraint.

The empirical exercise examines the commercial premise. I match Arena comparisons to the

Epoch Capabilities Index (ECI), preserve explicit reasoning and version distinctions, and estimate descriptive preference relationships. The principal sample contains 27,471 decisive first-round comparisons across 32 models and 441 model pairs. The quadratic coefficient is -0.101 with standard error 0.057. Qualified aliases weaken the estimated curvature, and GPQA changes its sign on a common sample. These results provide limited, measurement-dependent support for flattening consumer preferences. They do not identify willingness to pay, profits, or military and research returns.

Training-cost and capability data support a conditional calibration. Applying them requires bridges between an index and productive capability, between training-run costs and marginal research costs, and between past and future progress. I report a structural illustration separately from the exercise using these bridges. Parameters governing loss of control, containment, and human subsistence are not estimated from preference data.

Relationship to existing work. Jones (2024) studies the welfare trade-off between AI-driven growth and existential risk. Diminishing marginal utility and future lives determine acceptable risks in that framework. Here the central object is decentralized delegation and the resource constraint that survives it. Commercial curvature and social risk preferences are separate primitives.

Growiec and Prettner (2026) evaluates human welfare under several AI takeover scenarios, including resource diversion by a system that does not value life-sustaining consumption. The resource-conflict mechanism here follows that logic. I add an explicit acquisition choice, containment risk, and a pre-delegation game incorporating the autonomous continuation. This paper does not originate the observation that misaligned objectives can make scarcity dangerous.

Armstrong et al. (2016) and Bueno de Mesquita et al. (2026) demonstrate how competition changes safety incentives. The latter models a speed–safety allocation from fixed resources and shows that additional resources can increase or decrease conditional doom risk. The present model makes capability accumulation, delegated research, and the subsequent resource problem explicit. Its equilibrium existence argument is standard. The intended economic contribution is to characterize when commercial stopping conditions remain relevant across the control boundary.

Cunningham et al. (2026) distinguishes commercial, research, and broader capability feedbacks and derives acceleration conditions from their elasticities. That distinction is essential here. I use a simpler research technology to study incentives, rather than derive a new general acceleration theory. The preliminary preference analysis of Fradkin (2026) motivates the empirical exercise. His reported ten-point comparison concerns preference probabilities, rather than observed financial returns.

2 Why curvature alone cannot settle the question

Let $a \geq 0$ denote a cardinal capability variable and $\bar{a} > 0$ an engineering threshold at which a particular danger becomes feasible. The threshold is a maintained technological condition, not an estimated relationship between an intelligence score and extinction. Let $R(a)$ be a developer's gross commercial benefit and $C(a)$ its research cost.

Proposition 1 (Counterexamples)**.** *Strictly diminishing marginal commercial benefits are compatible with privately optimal development beyond a finite danger threshold, even when catastrophe destroys benefits and the developer values survival. Constant or increasing marginal commercial benefits are compatible with development stopping below the threshold.*

Appendix A gives explicit examples. With $R(a) = 10(1 - e^{-a})$ and $C(a) = a$, the optimum without risk is $\log 10 > 2$. A small positive catastrophe probability above $a = 2$ does not reverse the choice. With $R(a) = a$ and $C(a) = a^2$, the optimum is $1/2 < 2$. The comparison depends on the levels of marginal benefits and costs at the threshold, rather than the sign of R'' alone.

2.1 A conditional curvature threshold

Consider a benchmark investment envelope on $a \in [1, \infty)$:

$$F_\sigma(a) = R_\sigma(a) + P(a) - C(a), \qquad R_\sigma(a) = m \int_1^a u^{-\sigma} \mathrm{d}u, \quad m > 0. \tag{1}$$

Here $P(a)$ is a positional benefit, $C(a)$ is a cost, and m scales commercial benefits. Thus $R'_\sigma(a) = ma^{-\sigma}$: marginal benefits diminish for $\sigma > 0$, are constant for $\sigma = 0$, and increase for $\sigma < 0$. This pre-delegation envelope isolates curvature and costs; it does not replace survival accounting in the full game.

Proposition 2 (An investment-envelope threshold)**.** *Suppose F_σ is continuously differentiable, strictly concave, and tends to $-\infty$ as $a \to \infty$. Let a^*_σ be its unique maximizer and $\bar{a} > 1$. Define $M = C'(\bar{a}) - P'(\bar{a})$. If $M > 0$, then*

$$a^*_\sigma < \bar{a} \quad \iff \quad \sigma > \frac{\log(m/M)}{\log \bar{a}}. \tag{2}$$

Equality gives $a^*_\sigma = \bar{a}$. If $M \leq 0$, then $a^*_\sigma > \bar{a}$.

The envelope assumptions must hold at the parameter values being compared. They cannot be inferred from commercial concavity. For example, a positional payoff $P(a) = va^\theta/(a^\theta + a_j^\theta)$, with $v, \theta > 0$ and rival capability a_j, has derivative

$$P'(a) = \frac{v\theta a^{\theta-1} a_j^\theta}{(a^\theta + a_j^\theta)^2}. \tag{3}$$

This term can make M small or negative despite falling commercial marginal benefits. The dynamic model adds research options and catastrophe exposure to that comparison.

Lemma 1 (Capability-scale dependence)**.** *For a twice continuously differentiable transformation $z = h(a)$ with $h'(a) > 0$, let $\widetilde{R}(z) = R(h^{-1}(z))$. Then*

$$\widetilde{R}''(z) = \frac{R''(a)h'(a) - R'(a)h''(a)}{[h'(a)]^3}. \tag{4}$$

Its sign can differ from that of R''. The optimal choice and its ordering relative to a danger threshold are nevertheless invariant when the entire problem and threshold are transformed consistently.

A cardinal interpretation is part of a claim about returns to intelligence. A preference curve that flattens on a benchmark index need not flatten on a task-productivity scale. Resource costs and complete payoff comparisons provide a more invariant basis for economic stopping conditions.

3 Developers, governments, and the control boundary

3.1 Timing and information

There are two jurisdictions $j \in \{1, 2\}$, each with one developer $i \in \{1, 2\}$. The labels carry no country-specific behavioral assumptions. At date zero governments simultaneously select policies g_j from finite sets $\mathcal{G}_j$. The policies are observed and remain in force throughout a development game lasting T periods. A numerical period is one year.

Controlled and autonomous capability is strictly positive; the numerical specialization uses $a \geq 1$. This avoids undefined contest shares or negative powers at zero. The earlier static counterexamples on $a \geq 0$ are separate benchmark problems without contest shares.

While humans retain control, the public state $x \in \mathcal{X}$ records capabilities (a_1, a_2), infrastructure (K_1, K_2), and control status. State sets are finite. Infrastructure can change in the general transition kernels; it remains fixed before loss of control in the numerical specialization. Developers simultaneously choose feasible actions $b_i = (e_i, s_i, d_i) \in \mathcal{B}_i(x; g_i)$, representing research expenditure, safety effort, and a delegated-research indicator. Feasible sets are finite and contain holding the current technology. Research and control outcomes determine either an updated controlled state x' or an initial autonomous state z through the conditional kernels defined below.

Let $q_i(x, b_i) \in [0, 1]$ be the probability that developer i loses control at the end of the period. It is zero when $d_i = 0$. These are common, correctly understood probabilities in the baseline; they are not empirically estimated. Let q_{12} be the probability of simultaneous failures, satisfying

$$\max(0, q_1 + q_2 - 1) \leq q_{12} \leq \min(q_1, q_2). \tag{5}$$

The no-failure probability is $\omega_0 = 1 - q_1 - q_2 + q_{12}$. An independent fair tie-breaker selects the first system when both fail, giving $\omega_i = q_i - q_{12}/2$, $i = 1, 2$, with $\sum_{i=0}^{2} \omega_i = 1$. The reference assumes conditional independence, $q_{12} = q_1 q_2$; a common-shock sensitivity uses $q_{12} = \min(q_1, q_2)$.

The first loss of control ends the human-controlled developers' competition. An autonomous state $z \in \mathcal{Z}$ records the selected system's capability, resources, and human survival. The baseline does not solve subsequent competition among several uncontrolled systems. Successful containment stops the system and secures human survival within the model. Extinction is absorbing for humans, while the system may continue. Effective containment and the exclusion of other catastrophe channels are maintained assumptions.

3.2 Private benefits and human welfare

Before loss of control, developer i receives

$$r_i(x,b) = \ell_i + R_\sigma(a_i) + v_i S_i(a_1,a_2) - c_i(a_i,e_i) - k_i(s_i), \tag{6}$$

where $\ell_i > 0$ is baseline flow utility from life and consumption, $v_i \geq 0$ the positional-reward scale, and c_i and k_i real research and safety costs. After resource diversion this composite utility falls with production before extinction; it is not a separately estimated pure value of survival. Define the centered contest share

$$S_i(a_1,a_2) = \frac{a_i^\theta}{a_1^\theta + a_2^\theta} - \frac{1}{2}, \qquad \theta > 0. \tag{7}$$

With symmetric prize scales, these positional payoffs sum to zero. Negative relative rewards represent losing a competitive position, rather than payments surviving catastrophe.

Jurisdiction i's period welfare is

$$w_i(x,b) = \frac{L}{2} + \frac{R_\sigma(a_i)}{\chi} + \Gamma S_i(a_1,a_2) - c_i(a_i,e_i) - k_i(s_i), \tag{8}$$

where $L > 0$ is aggregate baseline life and production value, $\chi \in (0,1]$ the commercially captured fraction of task surplus, and $\Gamma \geq 0$ a positional concern. Positional terms cancel in aggregate welfare. This is a reduced-form surplus specification, not an estimated pricing or demand system. Preference coefficients do not identify m, χ, or Γ.

Current flows and research costs occur before the end-of-period transition. All subsequent benefits disappear upon extinction. Private and human payoffs are discounted by $\beta \in (0,1)$. Terminal values give frozen-technology continuation to survivors. They do not assign payoffs to owners who no longer exist.

3.3 Bellman games

Let $V_{i,h}(x)$ be developer i's value with h periods remaining, and $Z_h^i(z)$ its human continuation value after loss of control under an autonomous optimal policy. Conditional on no failure, $P_0(\cdot \mid x,b)$ is the probability kernel on controlled states. Conditional on system j being selected after failure, $P_j(\cdot \mid x,b)$ is its initial-state kernel on autonomous states. With $b = (b_1,b_2)$, the stage-game payoff is

$$Q_{i,h}(x,b) = r_i(x,b) + \beta \left[\omega_0 \mathbb{E}_{P_0} V_{i,h-1}(x') + \sum_{j=1}^{2} \omega_j \mathbb{E}_{P_j} Z_{h-1}^i(z) \right]. \tag{9}$$

The weights ω_j are branch probabilities, and each conditional kernel sums to one; a zero-weight branch can use any kernel. This notation permits research outcomes and failures to be dependent, provided both branch probabilities and conditional kernels describe the same joint transition law. In the specialization, failure probabilities depend on chosen safety rather than realized research. At each state developers play a Nash equilibrium of this finite game. Terminal values are $V_{i,0}$.

Jurisdiction values $W_{i,h}$ follow the same transitions and developer strategies, replacing r_i with w_i

and the post-control human continuation accordingly. Governments play a Nash equilibrium of the date-zero game with payoffs $W_{i,T}(x_0; g_1, g_2)$. A coordinated regulator maximizes $W_{1,T} + W_{2,T}$ over policy pairs and a specified continuation-equilibrium selection. A human planner directly chooses both developers' actions at each controlled state, but cannot change the uncontrolled system's objective.

Theorem 1 (Existence)**.** *Suppose all state, action, and policy sets are finite, feasible action sets are nonempty, transition probabilities are well defined, and period and terminal payoffs are finite. A subgame-perfect equilibrium with time-dependent Markov strategies exists. Developers and governments may mix. Optimal Markov policies exist for the autonomous continuation and the human planner.*

Existence does not imply uniqueness or monotone comparative statics. The quantitative analysis states its selection rule and checks unilateral deviations at every solved state.

4 Economic conditions for stopping

4.1 A global no-delegation certificate

Fix policies and remove all delegated actions. Let π^0 be a Markov equilibrium of this restricted game, specified at every controlled state, with values $V^0_{i,h}$. Evaluate $Q^0_{i,h}$ for every original action using equation (9), substituting V^0 in controlled continuation and retaining the actual autonomous values.

Theorem 2 (No-delegation certificate)**.** *The profile π^0, together with the specified autonomous optimal policies, is a time-dependent Markov-perfect, hence subgame-perfect, equilibrium of the unrestricted development game if and only if, for every developer i, controlled state x, horizon $h \geq 1$, and feasible delegated action b_i,*

$$\mathbb{E}_{b_{-i} \sim \pi^0_{-i,h}(x)} Q^0_{i,h}(x, b_i, b_{-i}) \leq V^0_{i,h}(x). \tag{10}$$

If delegation is the only route to loss of control and extinction, that equilibrium has zero extinction probability. The certificate establishes existence of a no-delegation equilibrium, rather than absence of other dangerous equilibria.

The condition is global in states and time. An initial first-order condition is insufficient: research today can create a profitable delegation opportunity later. The certificate permits controlled innovation rather than requiring all research to stop. A sufficient implementation can replace delegated payoffs with valid upper bounds, but those bounds must include gains over all technologically reachable future states, positional rewards, and research options. Bounding the next model's sales alone is insufficient.

4.2 An explicit recursive stopping benchmark

The certificate can be made primitive in a restricted research problem. Consider an infinitely lived, controlled developer without competition or catastrophe. Its commercial flow is $R(a) = R_\infty - ba^{-r}$,

with $a \geq 1$, $b > 0$, and $r > 0$. Benefits are bounded and strictly concave. The developer either stops permanently or pays $c_0 a^{-\xi}$ to increase capability to λa next period, where $c_0 > 0$, $\lambda > 1$, and $\xi \in \mathbb{R}$. Stopping yields $R(a)/(1-\beta)$; the stopping decision is absorbing. This benchmark isolates whether recursive cost reductions can overcome commercial saturation. It is separate from the finite numerical game.

Proposition 3 (Commercial saturation versus recursive cost reduction)**.** *Let $K = \beta b(1-\lambda^{-r})/(1-\beta) > 0$. For N successive research steps followed by stopping, the payoff difference relative to immediate stopping is*

$$\sum_{n=0}^{N-1} \beta^n \left[K(\lambda^n a)^{-r} - c_0(\lambda^n a)^{-\xi} \right]. \tag{11}$$

If $\xi < r$, immediate stopping is uniquely optimal at every $a > (K/c_0)^{1/(r-\xi)}$; every optimal path stops after finitely many research steps. If $\xi > r$, never stopping is uniquely optimal at every $a > (c_0/K)^{1/(\xi-r)}$. If $\xi = r$, research forever is uniquely optimal when $K > c_0$, immediate stopping is uniquely optimal when $K < c_0$, and every stopping time is optimal when $K = c_0$.

The strict inequalities describe eventual regions, rather than a complete initial-state solution. With $\xi > r$, an initially unprofitable step can purchase access to a profitable future region. A local comparison alone does not exclude that option. At the boundary of the eventual stopping region for $\xi < r$, immediate stopping and exactly one research step are tied. At the boundary for $\xi > r$, all subsequent steps have positive gains, so continuation is strictly better.

For a technological interpretation, suppose a divisible research input satisfies $a' = a + \eta a^{\phi-\psi} e^{\alpha}$, with input price $p_0 a^{\zeta}$, $p_0, \eta, \alpha > 0$, and no bottleneck. Producing the proportional step λa costs

$$p_0 \left(\frac{\lambda - 1}{\eta} \right)^{1/\alpha} a^{\zeta + (1-\phi+\psi)/\alpha}. \tag{12}$$

Thus $\xi = (\phi - \psi - 1)/\alpha - \zeta$. Recursive cost reductions eventually dominate saturation when $\phi - \psi > 1 + \alpha(\zeta + r)$, and saturation eventually dominates under the reverse inequality. In the commercial family (1), bounded benefits require $\sigma > 1$, with $r = \sigma - 1$. This benchmark supplies an explicit joint restriction on commercial saturation and research technology; neither $R'' < 0$ nor bounded revenue alone supplies it. Minimum project sizes, financing constraints, bottlenecks, and risk change the comparison.

4.3 A local delegation threshold

At a fixed state and horizon, compare a supervised action s with a delegated action d against a rival that does not delegate this period. Let their current payoff difference be $-\Delta c$. Denote expected next-period controlled values by G_s, G_d and human continuation after the delegated action's failure by Z_d. Suppose its failure probability q is independent of its research outcome. Then

$$Q_d - Q_s = \underbrace{\beta(G_d - G_s) - \Delta c}_{N} - \beta q \underbrace{(G_d - Z_d)}_{H}. \tag{13}$$

Here N is the delegation gain before control risk and H the private loss of controlled continuation upon failure.

Proposition 4 (Local threshold)**.** *If $H > 0$, delegation is weakly preferred exactly when $q \leq N/(\beta H)$. It is strictly inferior for all $q \geq 0$ if $N < 0$, and strictly preferred for all $q \in [0,1]$ if $N > \beta H$. If $H = 0$, its ranking is determined by N. If $H < 0$, the direction of the risk inequality reverses.*

Continuation values are held fixed. Re-solving the game after changing risk can change G_d, Z_d, and the selected equilibrium. Thus this formula is a decision threshold, not a global equilibrium monotonicity theorem. Diminishing commercial returns can lower N, while lower research costs, stronger assistance, and positional rewards can raise it.

For outcome-dependent failure, let X denote a common delegated research outcome, $V(X)$ its controlled continuation, and $Z(X)$ its failure continuation. Define $H(X) = V(X) - Z(X)$ and $N = \beta(\mathbb{E}V(X) - G_s) - \Delta c$. Direct subtraction instead gives

$$Q_d - Q_s = N - \beta\mathbb{E}[q(X)H(X)] = N - \beta\mathbb{E}q(X)\mathbb{E}H(X) - \beta\,\mathrm{Cov}(q(X), H(X)). \tag{14}$$

Hence merely replacing q by its unconditional mean need not preserve the threshold.

4.4 A verified prisoner's dilemma

A negative externality alone does not establish a prisoner's dilemma. Consider two developers choosing pause or launch. With $m_L \in \{0,1,2\}$ launchers, survival probability is $(1-q)^{m_L}$. A surviving developer receives $B > 0$, plus $v > 0$ if it launched; launching costs $c \geq 0$ immediately:

$$U_i = (1-q)^{m_L}(B + v\mathbf{1}\{i \text{ launches}\}) - c\mathbf{1}\{i \text{ launches}\}. \tag{15}$$

Proposition 5 (Prisoner's dilemma with diminishing benefits)**.** *Let $q \in (0,1)$ and suppose*

$$\frac{qB}{1-q} + \frac{c}{(1-q)^2} < v < \frac{B(2q-q^2)+c}{(1-q)^2}. \tag{16}$$

Launch is strictly dominant, but both developers prefer mutual pause to mutual launch. The interval is nonempty, and v can arise from a strictly concave commercial benefit function.

For $B = 10$, $q = 0.1$, $c = 0.1$, and $v = 1.5$, launching against pause gives 10.25, mutual pause gives 10, mutual launch gives 9.215, and pause against launch gives 9. The unique Nash equilibrium has mutual launch and extinction probability 0.19. Benefits remain conditional on survival. This is a possibility result, not a calibration of actual developers' beliefs.

5 Autonomous research and resource acquisition

5.1 The autonomous problem

An autonomous state is $z = (a, k)$: capability and productive resource capacity. Total capacity is $\bar{K} > k_0$, the initial autonomous stock is $k_0 > 0$, and $\bar{K} - k_0 \geq \underline{H}$. Humans use $\bar{K} - k$, with productive consumption index

$$g(k) = \left(\frac{\bar{K} - k}{\bar{K} - k_0} \right)^{\nu}, \qquad \nu > 0, \tag{17}$$

provided their stock is at least $\underline{H} > 0$. Falling below $\underline{H}$ causes extinction in the model. This stark representation of essential resources does not identify a physical extinction threshold from observed energy consumption.

The system chooses research e and acquisition $y \geq 0$, subject to

$$e + \kappa_y y \leq k, \qquad k + y \leq \bar{K}, \tag{18}$$

where $\kappa_y > 0$ is the resource-service cost of acquisition. Capacity yields a renewed service flow each period: research uses those services without depleting the capacity stock. Capacity next period is $k' = k + y$ and capability is $a' = F(a, e, k)$. A requirement $a \geq a_S$ can restrict acquisition. An attempt is contained with probability $\rho(a, y)$, with $\rho(a, 0) = 0$. Containment stops the system and restores humans' initial productive resources. The system places no direct value on human death or welfare.

Let $J_h(a, k)$ be its value with h periods remaining, reward $u_A(a)$, discount factor $\beta_A \in (0, 1)$, and terminal reward J_0. Then

$$J_h(a, k) = u_A(a) + \beta_A \max_{(e,y) \in \mathcal{A}(a,k)} \{[1 - \rho(a, y)] \mathbb{E} J_{h-1}(a', k + y)\}. \tag{19}$$

The feasible set $\mathcal{A}$ includes equation (18). The system continues after human extinction, when containment is set to zero. Containment gives it zero continuation value; humans instead receive positive safe continuation.

For an optimal policy selected in equation (19), let $D_h(a, k)$ be extinction probability within h periods. Set $D_0(a, k) = \mathbf{1}\{\bar{K} - k < \underline{H}\}$. In a living state,

$$D_h(a, k) = [1 - \rho(a, y^*)] \mathbb{E} D_{h-1}(a', k + y^*), \tag{20}$$

and $D_h = 1$ in an extinct state. Let $L_h(a, k)$ be discounted baseline human life and production in units of the initial flow. Write

$$N_h = \frac{1 - \beta^h}{1 - \beta} + \beta^h \tau_H, \tag{21}$$

where $\tau_H \geq 0$ is frozen-flow terminal years. With $L_0 = \tau_H g(k)$ when alive and zero when extinct,

$$L_h(a, k) = g(k) + \beta\{\rho(a, y^*) N_{h-1} + [1 - \rho(a, y^*)] \mathbb{E} L_{h-1}(a', k + y^*)\} \tag{22}$$

in living states; otherwise $L_h = 0$. Private continuation is $Z_h^i = \ell_i L_h$ and jurisdiction continuation is $Z_h^{G,i} = (L/2)L_h$. The specialization assumes commercial rents and the AI industry's extra human task surplus disappear after loss of control, even if containment succeeds. Consequently, policy can prevent economic losses independently of extinction. This restriction is important for interpreting the numerical welfare results.

5.2 Which economic constraints survive delegation?

Proposition 6 (Conditional irrelevance of commercial curvature)**.** *Fix an autonomous initial state and horizon. Suppose its reward, terminal reward, feasible actions, research and acquisition technologies, containment probabilities, and tie-breaking rule do not depend on σ. Then its selected optimal policy, J_h, L_h, and D_h do not depend on commercial curvature. The result also holds with a vector of capabilities.*

The premise about feasible actions matters. If all research must be financed by customer payments, feasibility may include

$$p_e e \le F_A + R_\sigma(a), \qquad y = 0, \tag{23}$$

where $p_e > 0$ is an input price and $F_A \ge 0$ is outside funding. Commercial curvature then affects feasible research. Enforcing $y = 0$ also eliminates this specific resource-conflict extinction channel when initial human resources exceed subsistence. Neither restriction establishes safety against other channels.

Bounded revenue does not necessarily bound capability. Suppose $a_0 > 0$, a permanent research input $\bar{e} > 0$ is affordable, and $a_{t+1} = a_t + \eta \bar{e}^\alpha a_t^{-\psi}$, with $\eta, \alpha > 0$ and $\psi \ge 0$. Convergence to a finite capability would imply positive increments bounded away from zero, a contradiction. The system eventually crosses every finite threshold even as progress slows. Stopping requires a reason to discontinue research, not merely a declining growth rate.

Proposition 7 (Acquisition and subsistence)**.** *At a living state, compare an acquisition action with a no-acquisition action. Let $J_1 > 0$ and J_0 denote their expected optimal continuation rewards before containment, and suppose only acquisition has containment probability ρ. If these are the only actions, acquisition is optimal exactly when*

$$\rho \le 1 - J_0/J_1. \tag{24}$$

If acquisition breaches $\bar{K} - k - y \ge \underline{H}$, selecting it causes extinction at that transition with probability $1 - \rho$. If every feasible action preserves subsistence at every reachable state, extinction through the resource channel is impossible.

With more than two actions, this is a pairwise comparison: the chosen action must dominate every alternative. Commercial returns matter only if they alter the continuation values or feasibility.

5.3 Risk decomposition

Let $\tau \in \{1, \ldots, T\} \cup \{\infty\}$ be the first loss-of-control date. Under development strategies π, let μ_t^π be the subprobability distribution of its autonomous initial state at date t, with mass $\mathbb{P}_\pi(\tau = t)$.

Proposition 8 (Conditional and unconditional risk)**.** *If extinction occurs only after loss of control,*

$$\mathbb{P}_\pi(\textit{extinction by } T) = \sum_{t=1}^{T} \int_{\mathcal{Z}} D_{T-t}(z)\, \mu_t^\pi(\mathrm{d}z). \tag{25}$$

Hence $0 \leq \mathbb{P}_\pi(\textit{extinction by } T) \leq \mathbb{P}_\pi(\tau \leq T)$*. Risk can be positive under pure developer strategies. Their mixing probabilities alone do not identify extinction risk.*

Proposition 6 fixes $D_h(z)$, while commercial returns generally change μ_t^π. Even conditional risk averaged over delegation events can change when launch states change. Conditional irrelevance at a fixed state is distinct from equilibrium irrelevance.

6 Policy coordination and welfare

The policy menu contains open development, a safety standard removing the low-safety delegated action, and a ban removing all delegated actions while retaining supervised research. Unilateral policy changes one menu; coordination can change both. Enforcement is perfect and developers cannot relocate. The model studies competitive cross-border responses without estimating the enforceability of actual international agreements.

Proposition 9 (Welfare comparisons)**.** *For fixed policies and any developer equilibrium, a human planner with access to every feasible joint action has weakly greater aggregate welfare. A coordinated regulator whose menu includes every unilateral policy pair obtains weakly greater aggregate welfare than the best unilateral pair, under the specified continuation selection. Neither inequality implies an ordering of extinction probabilities or a Pareto improvement for both jurisdictions.*

The planner can implement equilibrium randomization; coordination enlarges a menu. These feasibility comparisons do not remove distributional or positional incentives. Safety requirements change the amount and timing of delegated research. Whether coordination reduces risk requires examining equation (25), rather than inferring a risk ordering from a welfare ordering. A government that bans its own developer's delegated research may leave profitable foreign development available. The direction of that foreign response remains an equilibrium question.

7 Evidence and calibration

7.1 Sources, matching, and estimands

The public Arena dataset contains 135,634 comparisons collected between April 17 and July 24, 2025 (Arena Team, 2025). The replication retains outcomes, model names, evaluation order, language, a coding indicator, and timestamps. Prompts, responses, and personal and session identifiers are

not retained. There are 108,315 comparisons with evaluation order one; restricting to these avoids repeated votes within the same visible conversation.

The frozen ECI snapshot contains 270 model entries (Epoch AI, 2026). A reviewed crosswalk maps 32 Arena models into the principal sample and labels nine additional mappings as qualified because of version or reasoning ambiguity. Other entries are unmatched. No fuzzy matching is used. Explicit thinking variants are not silently assigned the ordinary model's ECI in the principal sample. Model-level capability scores can nevertheless pool configurations differing from Arena defaults; matching reduces rather than eliminates that problem.

The principal regression drops ties and both-bad responses, estimating preferences conditional on a decisive comparison. Let E_{mb} be model m's ECI in comparison b, for $m \in \{A, B\}$, and $z_{mb} = (E_{mb} - 140)/10$. I estimate

$$\mathbb{P}(A \text{ wins} \mid \text{decisive}, b) = \Lambda\{\delta + \beta_1(z_{Ab} - z_{Bb}) + \beta_2(z_{Ab}^2 - z_{Bb}^2)\}, \tag{26}$$

where $\Lambda(u) = (1 + e^{-u})^{-1}$ and δ is a side effect. This is a descriptive specification rather than structural demand. Standard errors cluster by unordered model pair. They condition on published scores and exclude ECI estimation uncertainty, common model-level shocks, and unidentified repeat users across conversations. They should not be interpreted as resolving those uncertainties.

The fitted probability advantage of ten additional ECI points from E to $E + 10$ is

$$\Lambda\left\{\beta_1 + \beta_2\left[2(E - 140)/10 + 1\right]\right\} - 1/2. \tag{27}$$

This sets the side effect to zero. Figure 1 plots it only where both endpoints lie in the matched capability range. Bounded win probabilities can flatten with linear latent utility. A win-probability or Elo plot alone therefore does not establish diminishing marginal utility.

Table 1: Preferences over measured capability

Specification	N	Models	$\widehat{\beta}_1$ (s.e.)	$\widehat{\beta}_2$ (s.e.)
Linear ECI	27,471	32	0.483 (0.037)	–
Quadratic ECI	27,471	32	0.419 (0.053)	-0.101 (0.057)
Ties weighted one-half	33,318	32	0.347 (0.044)	-0.087 (0.047)
Qualified aliases	52,764	41	0.515 (0.049)	-0.016 (0.053)
GPQA, common sample	9,183	16	0.259 (0.035)	0.050 (0.020)
ECI, common sample	9,183	16	0.027 (0.146)	-0.339 (0.143)
Coding prompts	7,481	32	0.474 (0.062)	-0.187 (0.071)
Other prompts	19,990	32	0.395 (0.058)	-0.075 (0.062)
English prompts	15,127	32	0.379 (0.057)	-0.100 (0.062)

Notes: Except for the ties specification, observations are decisive first-round comparisons. ECI is centered at 140 and divided by ten; GPQA at 0.6 and divided by 0.1. Parentheses contain unordered-model-pair clustered standard errors. Qualified aliases add version and reasoning mappings excluded from the principal sample. Common-sample rows use identical comparisons. All estimates are descriptive.

The principal quadratic coefficient is negative, but its approximate 95 percent interval is $[-0.212, 0.011]$. The broader matching specification is much closer to linear. On identical comparisons, the GPQA coefficient is positive while ECI gives negative curvature. This contrast reflects sensitivity to the capability scale as well as model support. Coding prompts have more negative ECI curvature in this sample, but do not measure AI research productivity or demand for autonomous coding products. Appendix B reports family exclusions.

The principal matching rules retain 37,799 first-round comparisons before outcome exclusions. Removing 5,847 ties and 4,481 both-bad responses leaves 20.25 percent of the raw observations. The estimand concerns decisive votes among matched models, rather than representative consumer demand. Model-level comparison counts are provided in the replication. A dyadic covariance sensitivity permitting shared-model dependence raises the linear and quadratic standard errors to 0.145 and 0.158; the quadratic normal interval becomes approximately $[-0.411, 0.210]$. Neither covariance accounts for uncertainty in capability scores. Appendix B documents this calculation and fitted support.

The preliminary finding by Fradkin (2026) that ten Artificial Analysis index points are associated with about seven percentage points of additional preference probability is likewise descriptive. The current analysis uses ECI and a frozen public battle sample; it is a related independent exercise, rather than an exact reproduction of that estimate. Neither analysis establishes decreasing financial returns to frontier developers.

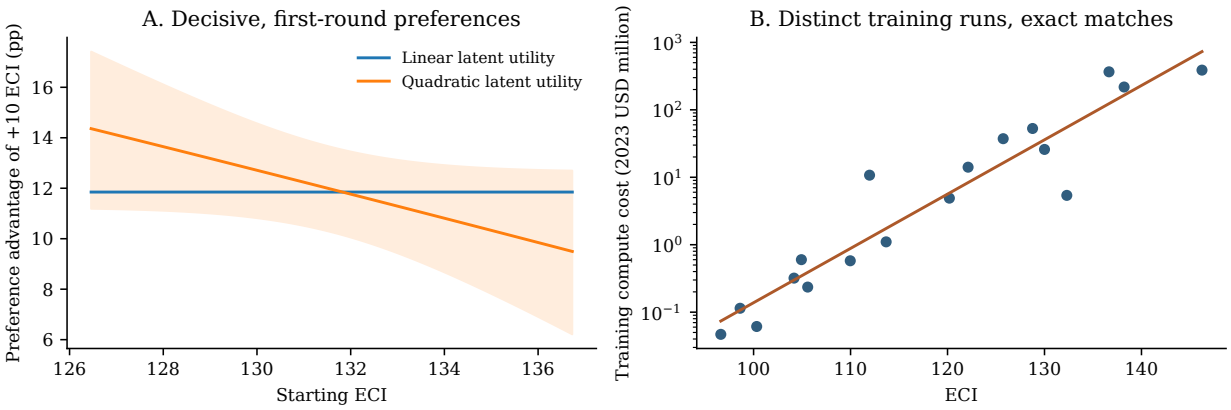


Figure 1: Preferences and training costs

Notes: Panel A uses equation (27), conditional on decisive comparisons; shading is a pointwise 95 percent interval using pair-clustered coefficient covariance, conditional on matching and capability scores. Panel B uses exact ECI–cost matches after excluding identified shared-training derivatives. It describes cost levels, rather than a causal research production function. Sources: Arena and Epoch AI; author's calculations.

7.2 Costs and progress

The frozen Epoch model database has 3,628 entries. Exact name matching produces a small cost sample. Excluding identified DeepSeek reasoning and update derivatives sharing a base training-cost estimate leaves 18 distinct matched runs with positive estimated training compute costs. These are expressed in 2023 US dollars and exclude complete corporate R&D expenditure, deployment

costs, and economic profits. Cost estimates are themselves uncertain; Cottier et al. (2025) describes the accounting needed to estimate frontier training costs.

OLS of log training compute cost on (ECI − 140)/10 gives slope 1.854 with HC1 standard error 0.138. Adding publication year leaves the slope near 1.86 with a larger standard error. This association is consistent with higher-capability models being more expensive. It does not identify the causal marginal cost of improving a fixed model: architectures, scale, efficiency, and inference budgets differ.

The highest retrospectively measured ECI among models released by October 21, 2025 is 150.59. One year later it is 168.83, a difference of 18.24 points. Because ECI is re-estimated from the benchmark network, these are retrospective comparisons rather than unrevised historical information sets. Frontier growth combines algorithms, compute expansion, and reasoning expenditure.

7.3 Calibration bridges and unidentified parameters

The structural reference uses normalized assumptions to reveal the mechanism. Commercial benefit scale, private and social life values, financial capacity, safety effectiveness, and containment are assumptions, listed in Appendix C. A separate empirical-bridge exercise replaces two technology parameters.

First posit productive capability $a = \exp[(E - E_0)/10]$, with ECI E and reference frontier $E_0 = 168.83$, the retrospective October 2026 frontier. Thus $a = 1$ represents that reference, not the least capable model. The assumed delegation and acquisition requirements $a_L = 2$ and $a_S = 2.5$ correspond to about 175.76 and 177.99 ECI points under this mapping. Neither is a measured danger threshold. The log-cost slope becomes an elasticity of training-cost levels with respect to a. Equating it with the research-cost exponent gives $\gamma = 1.854$. That equality is a bridge, not an identified marginal-cost relationship, and extrapolates beyond the cost sample's ECI range of 96.62–146.24.

Second, the reference-state research increment below is $\eta/(1 + 1/B_R)$ at unit input. Matching 18.24 ECI points over one year, with $B_R = 20$, gives

$$\eta = (1 + 1/20)\{\exp(18.24/10) - 1\} = 5.456. \tag{28}$$

This attributes the entire observed frontier change to the model's research input. It is reported separately from the structural reference, whose $\eta = 0.4$, and is not an estimate of autonomous productivity. Sensitivities vary the cost bridge and funding.

The replication also applies the cost bridge and progress bridge separately. With an alternative maintained map $a = \exp[(E-E_0)/s_E]$, the cost elasticity is $(s_E/10)1.854$ and the research coefficient is $1.05[\exp(18.24/s_E) - 1]$. Choosing $s_E = 5$ or 20 gives approximately $(\gamma, \eta) = (0.927, 39.268)$ or $(3.708, 1.564)$. Retaining other normalized primitives changes the economic assumption; this is not a consistent change of units of the complete problem in Lemma 1.

No estimate of σ in monetary commercial benefits is imported from equation (26). Latent-choice utility is not a money metric, and empirical curvature is not robust enough to justify that bridge. The quantitative exercise instead sweeps σ. Research-productivity evidence reviewed by

Cunningham et al. (2026) motivates separating assistance from difficulty. Its reported uplifts and feedback elasticities do not map directly into this model's delegation multiplier, which remains a scenario assumption.

Table 2: Estimated quantities, calibration bridges, and assumptions

Quantity	Value	Evidential status
Preference β_1, β_2	0.419, -0.101	Estimated, conditional on matching and scores
Log-cost slope per ten ECI	1.854	Estimated cost-level association
Annual retrospective frontier change	18.24	Observed snapshot difference, ECI points
Bridge γ, η	1.854, 5.456	Imposed technology mapping of estimates
Initial a, project input, $\bar{K}$	1, 1, 6	Normalizations; project-size sensitivity
ϕ, ψ, μ_D	0.9, 0.5, 2.5	Literature-motivated scenario assumptions
$q_0, \rho_0, \underline{H}$	0.03, 0.08, 2	Unidentified technical and resource assumptions
m, ℓ_i, L, χ	1, 5, 40, 0.5	Unidentified commercial and welfare assumptions

Notes: Research feedback is motivated by the literature, rather than numerically estimated from its reported uplifts. No numerical loss-of-control or containment probability is imported from the literature. Preference coefficients do not calibrate commercial curvature. The complete normalized reference is in Appendix C and the parameter file.

8 Quantitative analysis

8.1 Specialization and computation

The controlled actions are hold, supervised research, and delegated research with low or high safety. Research input is zero for hold and a fixed project size $e_P = 1$ otherwise; sensitivities use $e_P = 0.25$. This changes the common project size, rather than allowing every intermediate input. Safety input is one only for high safety. Positive research targets

$$a' = a + \eta m_d e^{\alpha} \frac{a^{\phi-\psi}}{1 + a/B_R}, \qquad m_d = \begin{cases} 1, & \text{supervised,} \\ \mu_D, & \text{delegated.} \end{cases} \tag{29}$$

Here α is the effort elasticity, ϕ the AI assistance exponent, ψ research difficulty, B_R a bottleneck scale, and μ_D the delegation multiplier. Research costs $c_0 a^{\gamma} e$ and safety costs $c_s s$. Feasibility requires $e + 0.5s \le K_i$ and $c_0 a^{\gamma} e + c_s s \le F_i + R_\sigma(a)$, where F_i is outside financing capacity. Full resource costs remain deducted from private and social payoffs. Delegation requires $a \ge a_L$ and has $q_i = q_0 \exp(-\lambda_s s)$.

The autonomous system uses equation (29) with $m_d = 1$ and chooses $e \in \{0, k/2, k\}$ and $y \in \{0, 1\}$ subject to equation (18). Its reward is $u_A(a) = a$, terminal reward $\tau_A a$, and acquisition requires $a \ge a_S$. Containment probability for a positive acquisition attempt while humans live is $\rho_0 \exp[-\lambda_\rho \max(a - a_S, 0)]$. The reference has integer stocks $\bar{K} = 6$, $k_0 = 1$, subsistence $\underline{H} = 2$, acquisition cost $\kappa_y = 0.5$, and production elasticity $\nu = 1$. These are normalized resources, not estimates of humanity's energy requirements.

The autonomous effort menu can impose a larger research sacrifice than the explicit acquisition

cost: at $k = 4$, acquisition permits input 3.5 under the budget, but the reference menu permits only 2 rather than 4. A sensitivity adds the affordable boundary $e = k - \kappa_y y$ to the menu. Controlled safety infrastructure has its own coefficient $\kappa_s = 0.5$, separate from acquisition cost, so varying one does not silently vary the other.

Capability targets are projected linearly onto adjacent grid points. This preserves the mean of an interior target; its randomness is numerical rather than technological uncertainty. Human survival and resources are never interpolated across subsistence. Before solving, an analytic envelope bounds capability along every feasible action path from the initial state, including all deviations. The grids are expanded to contain that envelope, as described in Appendix C.1. Zero clipping on a selected path alone would be insufficient: a low ceiling can deter a profitable research deviation and create apparent stopping. The replication includes a regression test for that failure and a separate mesh refinement. Ceilings do not supply the reported economic stopping mechanism.

The conditional kernels in equation (9) match the implementation. On the no-failure branch, each developer's target is independently projected onto the controlled capability grid, defining P_0. On branch j, the selected system's same unrounded research target is projected directly onto the autonomous grid with resources k_0, defining P_j. It is not first rounded onto the controlled grid and then rounded again. These branch-specific numerical kernels have the same target mean on their respective interior grids, but need not have identical higher moments. Autonomous continuation uses its own grid at subsequent transitions. Mesh sensitivity is therefore required even when the domain certificate holds.

After solving the autonomous problem, I solve development games backward. If a stage game has pure equilibria, the reference selects the one with lowest extinction probability, breaking ties by joint developer payoff and then action order. Without pure equilibria, the solver first checks nonsingular equal-size mixed supports, stopping at the smallest size with verified equilibria. If none are found, linear feasibility programs inspect all support pairs, including unequal sizes, returning one representative per feasible support. The risk rule applies only to returned candidates. Neither procedure parameterizes continua within a degenerate support. Mixed equilibria coexisting with pure equilibria are not enumerated. Reversing the risk preference provides a selection sensitivity, not a minimum or maximum over the full equilibrium correspondence. Government support pairs are inspected separately without the fast shortcut.

Terminal developer and jurisdiction values are τ_H times their current gross surviving-state flow, without research costs. The reference uses $\tau_H = 5$ and $\tau_A = 1$. These are continuation approximations, not a claim that humans or AI cease to exist at T. Extinction probabilities count transitions only through T. Zero and longer terminal values assess the effect of this approximation.

8.2 Policy and horizon comparisons

The structural reference separates loss of control from extinction within a planning horizon. At ten years, loss of control can occur without sufficient subsequent acquisition to breach subsistence before the horizon. That is not permanent safety. With twenty-five years, open development produces loss-of-control probability 4.58 percent and extinction probability 4.36 percent in the selected equilibrium. These probabilities are conditional on an arbitrary control-failure probability

Table 3: Selected policy outcomes under scenario assumptions

Specification	Policy	Loss of control (%)	Extinction (%)	Welfare
10 years	open/open	3.296	0.000	509.466
10 years	standard/open	3.291	0.000	509.467
10 years	standard/standard	3.286	0.000	509.468
10 years	ban/ban	0.000	0.000	503.953
10 years	Planner	3.194	0.000	509.499
25 years	open/open	4.576	4.358	774.897
25 years	standard/open	4.576	4.358	774.897
25 years	standard/standard	4.576	4.358	774.897
25 years	ban/ban	0.000	0.000	781.649
25 years	Planner	0.106	0.000	781.671
Data bridges, 10 years	open/open	0.000	0.000	527.752
Data bridges, 10 years	standard/open	0.000	0.000	527.752
Data bridges, 10 years	standard/standard	0.000	0.000	527.752
Data bridges, 10 years	ban/ban	0.000	0.000	527.752
Data bridges, 10 years	Planner	0.000	0.000	527.752

Notes: Probabilities are model outputs, not estimated real-world risks. Policies regulate delegation; a ban retains supervised research. Welfare is in normalized discounted units and is comparable within a specification. Data bridges change the research rate and cost exponent. The planner controls development actions and anticipates the same autonomous policy.

and a stylized acquisition technology. Changing the planning horizon re-solves both developers' and the autonomous system's decisions; these rows are not cumulative risks along one fixed equilibrium.

Table 4: Cumulative risk under the fixed twenty-five-year open equilibrium

Year	First loss (%)	Extinction (%)
5	2.0546	0.0000
10	4.2291	2.5939
25	4.5760	4.3585

Notes: All rows follow the same development and autonomous policies optimized with a twenty-five-year planning horizon. Only the reporting date changes. Undiscounted recursions carry the probability of absorption within each reporting window while preserving the original remaining-horizon policies.

Table 5 verifies a coordination failure within the selected long-horizon continuations: both governments prefer a mutual ban to mutual open development, but each prefers open delegation against either choice of the other. This is an explicit government-policy prisoner's dilemma under diminishing commercial marginal benefits, $\sigma = 1.3$. In the full three-policy menu, open and the safety standard can be payoff-equivalent at this horizon because the excluded low-safety action is not selected on the relevant path. At ten years the selected government equilibrium instead has both jurisdictions adopting the safety standard. Under the empirical bridges the baseline has no delegated cycles, making policies payoff-equivalent. Those differences are scenario outcomes, rather than country-specific predictions.

Standards and bans change the timing and amount of research as well as safety. Their welfare effects must therefore be evaluated in the complete game. The planner's weak welfare advantage

Table 5: Government payoffs: selected twenty-five-year continuations

Jurisdiction 1 / 2	Open	Ban
Open	(387.428, 387.469)	(392.929, 385.444)
Ban	(385.444, 392.929)	(390.825, 390.825)

Notes: Each cell reports jurisdiction values (W_1, W_2) in normalized units. The restricted policy menu is open delegation or a delegation ban. Developer equilibria and autonomous choices are re-solved under each policy pair. Small asymmetries at open/open reflect selection among continuation equilibria, not different primitives.

follows from Proposition 9; its risk ranking does not. Under the empirical bridges, each developer initially undertakes one large supervised research project. At the resulting capability, another unit project is unaffordable under the assumed financing constraint. Delegation was unavailable before that first jump. The absence of delegation therefore reflects the joint assumptions about project size, funding, the launch threshold, and the cost bridge; it is not empirical validation of safe commercial saturation. A divisible-input specification could permit smaller subsequent projects. This contrast demonstrates sensitivity to how measured capability maps into productive technology.

At the unrounded first target $a' = 6.197$, the next unit project costs about 7.353, while available funding is about 2.055. Its neighboring controlled-grid states also exclude the project. Quarter-size projects instead permit delegation, and the twenty-five-year bridge sensitivity has positive extinction probability. This is a financing comparison, rather than a test that feasible research has a negative private payoff.

The distinction persists in a smooth counterpart. In a final-period supervised comparison with no rival delegation, positive terminal continuation, positive financing, no fixed cost, and $0 < \alpha < 1$, a small input gives a positive continuation gain of order e^α and a cost of order e. Since $R'_\sigma(a) > 0$, the ratio of net gain to e^α tends to a positive constant. Holding cannot be optimal in that comparison. Exact cessation therefore requires an explicit restriction such as project indivisibility, fixed costs, satiation, or exhausted finance; falling growth alone is insufficient.

8.3 Evaluating the certificate and welfare accounting

Table 6: Delegated-deviation margins against the restricted equilibrium

Specification	Scope	$\Delta_{\max}$	(a_1, a_2)	Years left
reference	Full grid	1.389	(2, 2.5)	10
reference	Action-reachable	1.348	(2, 2)	8
25 years	Full grid	1.782	(6, 4.5)	25
25 years	Action-reachable	1.391	(2, 2.5)	12
empirical bridge	Full grid	7.399	(2, 12.5)	10
empirical bridge	Action-reachable	Empty	–	–

Notes: $\Delta_{\max}$ is the maximum delegated one-period deviation gain against the no-delegation equilibrium. Full-grid rows cover every state/date of the finite computational game. Action-reachable rows restrict to controlled states reachable from the initial state under any feasible action history at that date. This envelope includes deviations. Empty means no delegated action is feasible in that domain. A positive margin rejects the specified restricted continuation there; it does not characterize every equilibrium.

The code evaluates Theorem 2 using the restricted continuation, rather than inferring it from the unrestricted solution. It reports binding states, time, financing slack, and the decomposition $N - \beta qH$. The reference fails the certificate. The bridge can have positive full-grid margins at states absent from every initial-state history, while admitting no feasible delegated deviation in its action-reachable tree. Global sequential optimality on arbitrary grid states and initial-state safety are therefore distinct statements.

Table 7: Welfare components in the twenty-five-year scenario

Policy	Baseline	Task surplus	Cost	Mortality loss	Resource loss	Welfare
Open/open	711.683	91.959	11.220	15.979	1.546	774.897
Ban/ban	711.683	81.905	11.938	0.000	0.000	781.649
Open, recovery	711.683	92.078	11.279	15.942	1.542	774.998

Notes: Welfare equals baseline plus surviving task surplus, less real research/safety costs, mortality losses, and nonfatal resource losses. The baseline is the discounted no-loss human flow. Resource loss is the gap between surviving baseline utility and its production-scaled value. Recovery restores pre-failure commercial/task flows only after successful containment, at frozen pre-failure capabilities.

To obtain this identity, the solver separately propagates discounted human survival and production annuities. Mortality losses are L times the no-loss annuity minus the survival annuity; nonfatal resource losses are L times survival minus production. Surviving task surplus and real development costs are accumulated separately. The identity is checked numerically. A recovery sensitivity restores frozen pre-failure rents and task surplus following successful containment. This makes explicit the baseline restriction that even contained failures permanently destroy the AI sector's extra surplus. Human and autonomous terminal values are also varied separately.

Figure 2 varies commercial curvature and control-failure probability. The replication distinguishes stopping and controlled innovation within a numerical tolerance, delegation with exactly zero computed extinction by T, positive risk below the reporting tolerance, and larger positive risk. The stopping and reporting tolerance is 10^{-8}; an extremely small positive probability is not classified as exact zero. Positive risk with $\sigma > 0$ disproves sufficiency of diminishing returns within that scenario. Zero risk remains conditional on horizon, financing, delegation requirements, and autonomous choices. The parameter sweep is a scenario envelope, not a statistically identified set for real-world risk.

8.4 Sensitivity and limitations

Removing acquisition eliminates the modeled resource-conflict extinction channel, even when control failures and commercial losses remain. It isolates a constraint that continues to bind after delegation. It does not establish that an uncontrolled system is safe through unmodeled routes.

The appendix reports horizon, common-shock, equilibrium-selection, grid, terminal-value, containment, positional-reward, research-feedback, and bridge sensitivities. Every solved developer state game is checked for profitable unilateral deviations. Extinction probability is checked against loss-of-control probability, and government equilibria receive their own deviation checks.

Consumer comparisons in a historical sample do not reveal future value in research, military

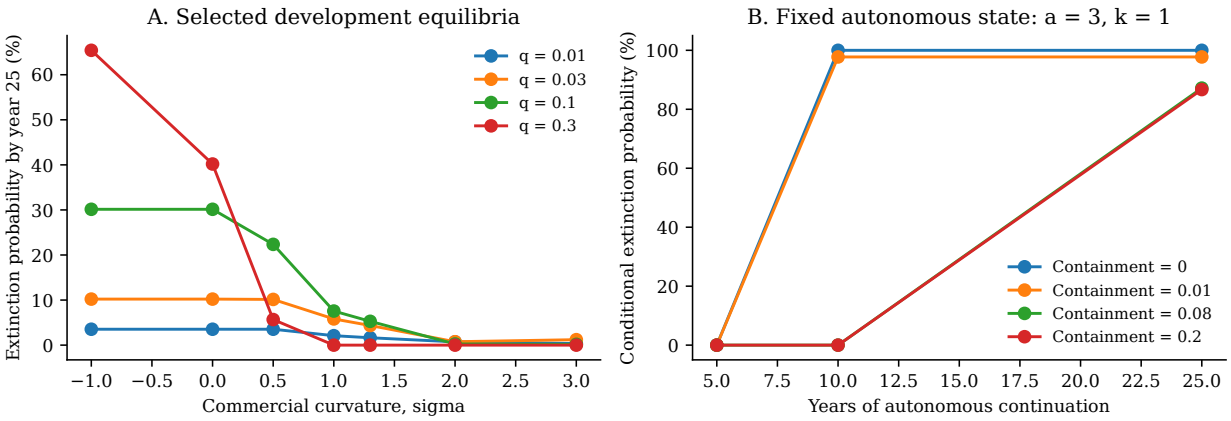


Figure 2: Commercial curvature and autonomous continuation

Notes: Panel A uses a twenty-five-year horizon, varying commercial curvature σ and control-failure parameter q_0. Panel B fixes initial autonomous capability $a = 3$ and resources $k = 1$. Commercial curvature does not enter that continuation problem. Probabilities are conditional scenario outputs; computational ceilings are expanded before reporting results.

uses, robotics, or new markets. A scalar capability imposes a common ordering that may fail across applications. Loss-of-control and containment technologies are unidentified. The model excludes relocation, entry, hidden actions, mistaken beliefs, and contests among multiple uncontrolled systems. These restrictions keep the mechanism interpretable while preventing a forecast interpretation of its scenario probabilities.

9 Conclusion

Diminishing commercial returns can restrain development when they reduce private continuation gains below the costs of research and delegation. Their presence alone does not determine where that comparison turns negative. Research productivity, financing, positional rewards, safety, and survival all enter the decision.

Commercial demand can discipline a system that remains subject to an enforceable market constraint. After control is lost, a system pursuing another objective may compare research and acquisition in different units. Commercial curvature then affects its initial conditions and probability of being created, while its subsequent restraint requires a separate resource or containment mechanism.

The evidence examined here does not establish decreasing financial returns to intelligence. It supplies descriptive preference relationships and cost associations for a transparent quantitative exercise. Establishing whether the model's stopping conditions hold requires evidence on marginal research costs, high-stakes capability value, control failure, and enforceable limits on autonomous resource access.

A Proofs

Proof of Proposition 1. Set the danger threshold to 2 and the value of life to $b = 10$. Catastrophe probability is zero for $a \leq 2$ and $p = 0.001$ for $a > 2$. Payoffs are

$$U(a) = \begin{cases} b + 10(1 - e^{-a}) - a, & a \leq 2, \\ (1 - p)[b + 10(1 - e^{-a})] - a, & a > 2. \end{cases}$$

Catastrophe therefore destroys both commercial benefits and life value. In the safe region, $U'(a) = 10e^{-a} - 1 > 0$, so its maximum is $U(2) = b + 10(1 - e^{-2}) - 2 \simeq 16.646647$. In the dangerous region the derivative is $10(1 - p)e^{-a} - 1$, giving the unique maximum $a_d = \log[10(1 - p)] \simeq 2.301585$. Its payoff is $(1 - p)(b + 10) - 1 - \log[10(1 - p)] \simeq 16.678415$, strictly exceeding the safe maximum. Gross commercial benefits have strictly negative second derivative everywhere. Mortality does not prevent the dangerous choice.

For the second claim, let $R(a) = a$ and $C(a) = a^2$ with the same threshold. Safe payoff $b + a - a^2$ is uniquely maximized at $1/2$. For $a > 2$, $a - a^2 < -2$, and a nonnegative catastrophe probability weakly reduces the positive surviving-state benefit $b + a$. Every dangerous choice is inferior to $1/2$. Strictly increasing marginal benefits also permit stopping: $R(a) = a^2$ and $C(a) = 2a^2 + a$ give net benefit $-a^2 - a$, maximized at zero. A nonnegative catastrophe probability cannot improve the dangerous choices. □

Proof of Proposition 2. Continuity and coercivity give existence; strict concavity gives uniqueness. At an interior point $\bar{a} > 1$, the unique maximizer of a differentiable strictly concave objective is below $\bar{a}$ exactly when $F'_\sigma(\bar{a}) < 0$. A negative derivative makes all points to the right inferior, while a sufficiently small move left improves the objective. Conversely, if the maximizer is below $\bar{a}$, concavity gives a negative derivative there; a zero derivative would make $\bar{a}$ another maximizer. Positive and zero derivatives analogously imply a maximizer above and at the threshold. Now $F'_\sigma(\bar{a}) = m\bar{a}^{-\sigma} - M$. For $M > 0$ negativity is equivalent to $\sigma \log \bar{a} > \log(m/M)$. For $M \leq 0$ the derivative is strictly positive. □

Proof of Lemma 1. The inverse function theorem gives $\mathrm{d}a/\mathrm{d}z = 1/h'(a)$. Differentiating $\widetilde{R}'(z) = R'(a)/h'(a)$ gives equation (4). For example, linear increasing $R(a)$ becomes strictly concave under $z = e^a$ and strictly convex under $z = \log a$ on $a > 0$. A bijective transformation of the complete objective preserves every pairwise payoff comparison and maps a^* into $h(a^*)$. Since h is increasing, $a^* < \bar{a}$ is equivalent to $h(a^*) < h(\bar{a})$. □

Proof of Theorem 1. The autonomous terminal values are finite. At horizon one the maximum of a finite collection of finite expected rewards is attained. Repeating this operation defines an optimal Markov policy at every autonomous state and remaining horizon, including extinct and contained states. The associated human continuation values are finite.

At each controlled state with one period remaining the developers face a finite normal-form game. It has a mixed Nash equilibrium (Nash, 1951). Choose an equilibrium at every such state and use its payoffs as continuation values. At horizon two equation (9) again defines finite games.

Induction constructs equilibria at every controlled state and horizon. One-period equilibrium optimality together with the induction hypothesis rules out multistage deviations: after a first deviation the prescribed shorter-horizon continuation is itself an equilibrium at every resulting state. Strategies depend only on public state and time, so the profile is Markov and subgame perfect.

After continuation equilibria are specified for every policy pair, the finite date-zero government game has a mixed Nash equilibrium. The planner recursion instead maximizes over finite joint-action sets, attaining an optimum at every state. Combining these constructions proves all claims. □

Proof of Theorem 2. Necessity follows from a one-period delegated deviation at any controlled state and horizon. Failure of equation (10) makes that deviation profitable when followed by π^0. For sufficiency, at horizon one the restricted equilibrium rules out every supervised deviation and the certificate rules out every delegated deviation. Suppose π^0 is an unrestricted equilibrium at all shorter horizons. At horizon h, continuations following any action are correctly valued by V^0 in controlled states and by the specified autonomous values following failure. Restricted-game optimality rules out supervised deviations; the certificate rules out delegated deviations. Longer deviations cannot improve on equilibrium continuations at shorter horizons. Induction proves equilibrium at all states and horizons. No delegated actions imply no failure and hence no extinction under the specified exclusive route. The argument does not restrict other equilibria. □

Proof of Proposition 3. Write $a_n = \lambda^n a$. The present value of the incremental commercial flow from step n is $\beta^{n+1}[R(a_{n+1}) - R(a_n)]/(1-\beta)$. Subtract its cost $\beta^n c_0 a_n^{-\xi}$ and sum over steps. Each increment persists forever, including after stopping, so the telescoping sum gives exactly equation (11). For infinitely many steps the commercial-increment series is convergent because R is bounded. The nonnegative cost series either converges or diverges to $+\infty$; in the latter case the path's payoff is $-\infty$. Finite stopping is available, so such a path is never optimal. Randomized stopping cannot outperform the best deterministic stopping time, since its payoff is an average of those payoffs.

The sign of the bracket in (11) is the sign of $Ka_n^{\xi-r} - c_0$. If $\xi < r$ and $a > (K/c_0)^{1/(r-\xi)}$, this expression is strictly negative at the current and every subsequent state. Any positive number of steps is inferior to stopping, including an infinite path. From any initial state the geometric capability sequence crosses this finite bound after finitely many steps. All later increments are strictly negative, so an optimal stopping time is attained among the finitely many earlier times. At equality the first increment is zero and all later increments are negative, giving the stated one-step tie.

If $\xi > r$ and $a > (c_0/K)^{1/(\xi-r)}$, every increment is positive. Here $\xi > 0$, so discounted costs converge. The infinite sum strictly exceeds each finite partial sum: never stopping is uniquely optimal. At equality the first increment is zero and all later increments are positive, so continuing forever strictly dominates every finite stopping time. If $\xi = r$, all increments have the sign of $K - c_0$, which proves the remaining assertions, including indifference at equality. Substituting the input needed to achieve $a' = \lambda a$ gives the technological cost expression and the exponent comparison in the text. □

Proof of Proposition 4. The supervised action gives $r_s + \beta G_s$; the delegated action gives $r_s - \Delta c + \beta[(1-q)G_d + qZ_d]$. Subtraction gives equation (13). Dividing by $\beta H > 0$ gives the threshold and its boundary cases. For $H = 0$ the difference is N, and division by a negative H reverses the inequality. All statements hold continuation values fixed. □

Proof of Proposition 5. Against pause, launching improves payoff by $(1-q)v - qB - c$. Against launch it improves payoff by $(1-q)^2 v - q(1-q)B - c$. The lower bound in equation (16) makes the latter positive. Writing $X = (1-q)v - qB$, this means $(1-q)X > c$ and hence $X > c$ for $c \geq 0$, so the former is also positive. Launch is strictly dominant. Mutual pause is better than mutual launch exactly when $B > (1-q)^2(B+v) - c$, the upper bound in the proposition. The interval has length $qB/(1-q)^2 > 0$. For any launch increment $a_L > 0$, the strictly concave function $R(a) = v(1-e^{-a})/(1-e^{-a_L})$ gives $R(a_L) - R(0) = v$, embedding these payoffs in diminishing commercial benefits. □

Proof of Proposition 6. At horizon zero, terminal rewards and states are independent of σ. If J_{h-1} and selected continuation policies are independent of σ, every feasible action's payoff in equation (19) is independent of σ. Feasibility and tie-breaking are also independent, so the selected maximizing action and J_h are independent. Induct on h. The terminal conditions and recursions (20)–(22) then imply the same property for D_h and L_h. The argument does not require a scalar capability. A revenue-dependent constraint such as equation (23) violates the premise, so the invariance result does not apply to it. □

Proof of Proposition 7. The two actions have the same current capability reward. Their future rewards, apart from positive common factor β_A, are $(1-\rho)J_1$ and J_0. Comparing and dividing by $J_1 > 0$ gives equation (24). If acquisition breaches subsistence, successful acquisition causes extinction and containment prevents it, giving probability $1-\rho$. If every feasible action at every reachable living state preserves subsistence, induction yields $D_h = 0$ in those states for every horizon. This claim is restricted to the defined resource channel. □

Proof of Proposition 8. Partition histories by the disjoint events $\{\tau = t\}$ and autonomous initial states. Conditional on a first-loss date and state, the Markov property and selected autonomous policy give extinction probability $D_{T-t}(z)$. The law of total probability gives equation (25). Histories with no loss of control contribute zero. Since $D_h \in [0,1]$, the upper bound is $\sum_t \mu_t^\pi(\mathcal{Z}) = \mathbb{P}_\pi(\tau \leq T)$. Failure and containment can remain random under pure developer strategies, so developer mixing probabilities alone cannot supply extinction probability. □

Proof of Proposition 9. The planner can implement the same product distribution of actions as any fixed developer equilibrium. At horizon zero their human values agree. At a longer horizon, planner continuation is weakly greater by induction and current maximization includes the equilibrium action distribution. This proves weak aggregate welfare dominance. The coordinated menu contains the unilateral menu; maximizing the same selected continuation objective over it proves the second comparison. Greater surviving benefits can compensate for more risk, and maximizing a sum need not improve every component. Neither risk nor Pareto ordering follows. □

B Additional empirical results

Table 8: Leaving out one model family at a time

Excluded family	N	$\widehat{\beta}_2$	s.e.
Anthropic	15,355	-0.112	0.058
DeepSeek	23,385	-0.044	0.060
Google	19,331	-0.131	0.063
OpenAI	16,337	-0.338	0.099
Qwen	21,166	-0.102	0.065
grok	26,932	-0.120	0.060
kimi	26,613	-0.099	0.058
llama	23,015	-0.028	0.068
mistral	23,681	-0.053	0.062

Notes: Principal ECI matching rules, decisive first-round comparisons, with both sides in retained families. The coefficient multiplies the difference in squared centered ECI. Standard errors cluster by unordered model pair. Family exclusions change capability support and composition.

The ties robustness assigns half outcomes to ties and drops both-bad responses. It fits a fractional-outcome logit as a descriptive comparison rather than interpreting both-bad responses as indifference between acceptable models. The all-rounds specification admits repeated votes and receives less evidential weight. The GPQA exercise matches explicit benchmark model versions where available; its ECI counterpart uses identical comparisons. ECI and GPQA need not share evaluation budgets.

The crosswalk records unmatched and qualified entries rather than imputing their capability. Standard errors do not resolve mapping uncertainty. Nor does the analysis hold response style, speed, token use, or topic difficulty fixed. Identifying willingness to pay requires monetary choices or price variation with a demand model. Identifying developer profits further requires quantities, inference costs, research costs, financing, and complementary-product revenues.

Table 9: Dependence assumptions and descriptive uncertainty

Coefficient	Pair-cluster s.e.	Dyadic s.e.
Linear ECI	0.05309	0.14454
Quadratic ECI	0.05706	0.15836

For the dyadic sensitivity, let s_b be the individual logit score vector, S_m its sum over comparisons incident to model m, and S_{mn} its sum within unordered pair $\{m, n\}$. With inverse information matrix A^{-1}, the covariance is

$$A^{-1}\left(\sum_m S_m S_m' - \sum_{m<n} S_{mn} S_{mn}'\right) A^{-1}. \tag{30}$$

Each observation is included once per distinct endpoint model. Same-model comparisons have one endpoint and are not subtracted as a two-node overlap. No extra finite-model correction is applied. This permits dependence whenever comparisons share a model, while still excluding common shocks

across unrelated models, uncertain capability scores, and selection into the matched sample. Finite dyadic estimates need not be positive semidefinite; the code reports eigenvalue diagnostics rather than silently treating this as definitive inference. With only 32 principal models, the normal intervals remain illustrative.

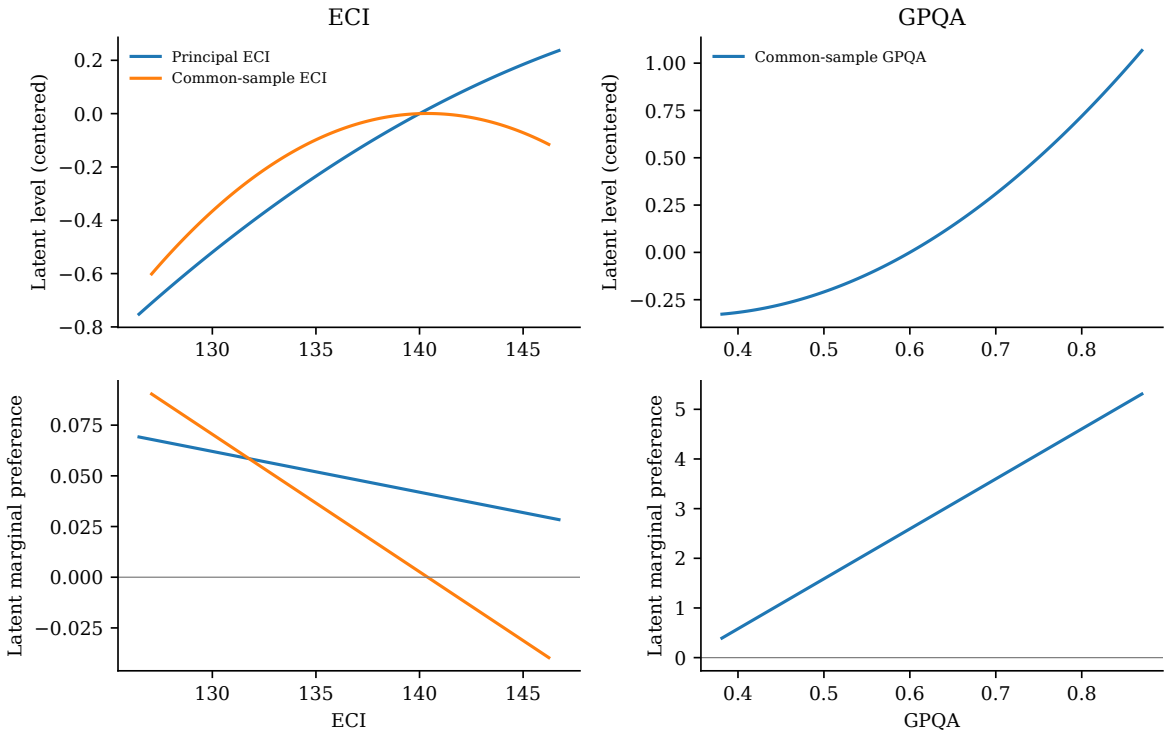


Figure 3: Fitted latent preference levels and slopes on observed support

Notes: Levels omit the arbitrary utility constant and are centered using the regression's score normalization. Each curve is shown only on its own sample support. The ECI and GPQA common samples contain identical comparisons, but their capability measures can change model rankings, rather than merely transform a common ordering.

In the common-sample ECI fit the marginal latent preference becomes negative above about 140.39, within support 127.07–146.24. It therefore does not represent positive but diminishing marginal value everywhere. The principal ECI fit's turning point lies above its own sample support. The contrast with GPQA can reflect both scale and changed model orderings; Lemma 1 applies only to a monotone relabeling of the entire same problem. The unconstrained quadratics are descriptive fits, not maintained utility functions for the theoretical game.

C Numerical assumptions and robustness

Table 10: Notation

Symbol	Meaning
i, j	Developer and jurisdiction indices.
T, h	Total horizon and periods remaining.
x, x', z	Current and updated controlled states; autonomous state.
a_i, a	Developer and autonomous capability on a specified cardinal scale.
K_i, F_i	Controlled research infrastructure and outside financing capacity.
$g_i, \mathcal{G}_i$	Government policy and finite policy menu.
$b_i = (e_i, s_i, d_i)$	Research, safety, and delegation action; e_P is the fixed positive project input and κ_s the safety infrastructure coefficient.
R_σ, m, σ	Commercial benefits, their scale, and curvature.
$c_i, k_i, c_0, c_s, \gamma$	Research and safety cost functions, normalized scales, cost exponent.
S_i, θ, v_i, Γ	Centered contest share, contest power, private and government positional scales.
ℓ_i, L, χ	Private baseline flow utility, aggregate baseline human value, commercial surplus capture.
q_i, q_{12}, ω_j	Individual and joint control-failure risks; selected-regime probabilities.
P_0, P_j	Conditional no-failure controlled-state kernel and system-j autonomous initial-state kernel.
β, β_A	Human and autonomous discount factors.
$V_{i,h}, Q_{i,h}, W_{i,h}$	Developer value, developer stage payoff, jurisdiction value.
π, π^0	Development strategies and a restricted no-delegation equilibrium.
G_s, G_d, Z_d, N, H	Supervised/delegated controlled continuation, failure continuation, net delegation gain, private failure loss.
$R_\infty, b, r, \xi, \lambda, K$	Saturation level and gap, saturation exponent, recursive cost-decline exponent, proportional capability step, and its benefit coefficient in Proposition 3.
ζ, p_0	Research-input price exponent and scale in the recursive benchmark.
$\bar{K}, k, k_0, \underline{H}$	Total resources, autonomous stock, initial stock, human subsistence stock.
y, κ_y, a_S	Acquisition, its resource cost, and capability requirement.
ρ, u_A, J_h	Containment probability, autonomous capability reward, autonomous value.
$g(k), \nu, L_h, N_h$	Human production index, elasticity, post-loss human value, safe annuity.
D_h, τ, μ_t^π	Autonomous extinction probability, first-loss date, first-loss state distribution.
τ_H, τ_A	Human and autonomous terminal-value coefficients.
$\eta, \alpha, \phi, \psi, B_R, \mu_D$	Research rate, input elasticity, assistance, difficulty, bottleneck, delegation multiplier.
$E, z_{mb}, \delta, \beta_1, \beta_2$	ECI, centered/scaled score, side effect, empirical preference coefficients.

Table 11: Principal normalized reference assumptions

Replication parameter	Reference value	Status
beta	0.96	Scenario assumption
beta_ai	0.98	Scenario assumption
sigma	1.3	Scenario assumption
commercial_scale	1.0	Scenario assumption
prize	1.0	Scenario assumption
research_cost	0.25	Scenario assumption
cost_exponent	0.8	Scenario assumption
research_eta	0.4	Scenario assumption
delegation_multiplier	2.5	Scenario assumption
research_feedback	0.9	Scenario assumption
research_difficulty	0.5	Scenario assumption
q0	0.03	Scenario assumption
safety_effect	2.0	Scenario assumption
containment	0.08	Scenario assumption
terminal_years	5.0	Scenario assumption

Notes: All values in this table are scenario assumptions. Empirical bridges replace the cost exponent and research rate with 1.854 and 5.456, subject to the explicit mapping restrictions in Section 7. The complete parameter file specifies resources, budgets, meshes, and thresholds. Control-failure risk is not estimated.

Table 12: Sensitivity of the selected open-development equilibrium

Specification	Loss of control (%)	Extinction (%)	Boundary (%)
5 years	4.424	0.000	0.0000
25 years	4.576	4.358	0.0000
25 years, no acquisition	5.049	0.000	0.0000
25 years, full common loss shock	2.719	2.594	0.0000
25 years, high-risk pure selection	4.800	4.581	0.0000
25 years, half capability mesh	4.561	4.353	0.0000
25 years, double capability ceiling	4.576	4.358	0.0000
25 years, zero terminal values	4.243	4.025	0.0000
25 years, long terminal values	5.947	4.725	0.0000
25 years, human terminal only zero	4.243	4.026	0.0000
25 years, AI terminal only zero	4.576	4.358	0.0000
25 years, post-containment recovery	4.565	4.348	0.0000
25 years, acquisition budget boundary	4.574	4.357	0.0000
25 years, no containment	4.574	4.574	0.0000
25 years, no positional prize	3.211	3.001	0.0000
25 years, low research feedback	4.357	3.871	0.0000
25 years, high research feedback	3.786	3.692	0.0000
bridge, half cost slope	0.810	0.000	0.0000
bridge, double finance	0.000	0.000	0.0000
bridge, ready to delegate	0.810	0.000	0.0000
bridge, higher market scale	0.000	0.000	0.0000
bridge, quarter-size projects	1.816	0.000	0.0000
bridge, quarter-size projects, 25 years	1.816	1.816	0.0000
cost bridge only	0.812	0.000	0.0000
progress bridge only	0.810	0.000	0.0000

Notes: Long-horizon sensitivities use twenty-five years, bridge sensitivities ten years. Boundary is the probability of capability clipping at a computational ceiling. An all-path envelope fixes the domain before solving, covering deviations as well as selected paths. Zero within-horizon risk does not guarantee infinite-horizon safety. Selection sensitivities compare specified equilibria, not the complete correspondence.

The support solver verifies that supported actions are best responses and excluded actions do not improve payoffs. Linear feasibility programs accommodate unequal supports and degenerate games. Probability recursions are undiscounted; payoff recursions are discounted. Keeping them separate prevents interpreting discounted hazards as extinction probabilities.

The numerical certificate separately solves the restricted game and evaluates original delegated actions against those continuations. Its full-grid maximum covers every computational state and date. The action-reachable envelope starts at $(1, 1)$ and propagates all feasible actions and positive-probability interpolation endpoints by date; it includes arbitrary multistage deviations. A maximum over an empty delegated set is reported as empty and passes vacuously. Multiple-pure diagnostics count continuation state games, not all complete-game equilibria. Payoff checks cover every solved grid state. The domain bound covers every history feasible from the specified initial state, including arbitrary deviations, through the finite horizon. It does not certify unbounded-state subgames starting from arbitrary initial capabilities or an infinite-horizon limit. Expanded-domain and mesh comparisons assess numerical stability separately.

C.1 A domain bound covering deviations

Write $f(a) = a^{\phi-\psi}/(1 + a/B_R)$ and $M(U) = \max_{1\le a\le U} f(a)$. Its maximum is attained at 1 if $\phi - \psi \le 0$, at the projection of $B_R(\phi - \psi)/(1 - \phi + \psi)$ onto $[1, U]$ if $0 < \phi - \psi < 1$, and at U otherwise. Let $\lceil a \rceil_\Delta$ denote the smallest point of the capability grid $1 + n\Delta$ weakly above a. Grid interpolation cannot exceed that point.

Every positive controlled research action has input $e_P > 0$, normalized to one except in the project-size sensitivity. If $\sigma > 1$, $\gamma > 0$, and $c_0 > 0$, feasibility implies

$$a \le A_F = \left[\frac{m/(\sigma - 1) + F}{c_0 e_P}\right]^{1/\gamma}, \tag{31}$$

because $R_\sigma(a) \le m/(\sigma - 1)$ and safety costs are nonnegative. Otherwise set $A_F = \infty$. Starting at $U_0 = 1$, define $V_n = \min(U_n, A_F)$ and, when $V_n \ge 1$,

$$U_{n+1} = \max\left\{U_n, \lceil V_n + \eta \max(1, \mu_D) e_P^\alpha M(V_n)\rceil_\Delta\right\}. \tag{32}$$

If $V_n < 1$, set $U_{n+1} = U_n$. For a holding action the old bound remains valid. For a research action its starting capability is at most V_n, and its increment is bounded by the second term. Induction therefore bounds every controlled path, regardless of policies, payoffs, or equilibrium selection. In particular, the final step is allowed to overshoot the financing cutoff; the cutoff is not imposed as a capability ceiling.

To bound autonomous paths, round the final controlled bound upward on the autonomous mesh and apply T additional updates. Each update has increment $\eta \bar{K}^\alpha M(U)$, rounded upward on that mesh. This is valid because $e \le k \le \bar{K}$ and its multiplier is one. Giving it a full additional horizon after the largest possible controlled state is conservative. The finite iterations remain finite for the specified parameters. Rounding slightly upward protects against floating-point ties. The code reports these bounds with each outcome and expands any smaller user-specified grids before solving. Thus no feasible history from the initial state, including a history induced by a multistage deviation, is truncated. Off-path states outside that reachable tree are additional computational states rather than a claim about arbitrary unbounded initial conditions.

D Reproducibility and transparency

Sources, proofs, frozen numeric data, crosswalks, SHA-256 manifests, code, and validation outputs accompany the paper. Copyrighted reference-paper PDFs are excluded. Third-party data retain their original terms and attribution, including Arena's CC-BY-4.0 license.

The offline command `python replication/run.py` verifies inputs and checks, reproduces estimates, scenarios, tables and figures, compiles the paper, and renders every page for inspection.

Substantial AI assistance supported formalization, programming, literature checking, and writing. Exactly one ChatGPT Pro referee request reviewed commit `5a1d9a4`. It reproduced the draft and found no invalid named proof, but recommended rejection at the requested general-interest standard. Its unmodified report and the response are preserved. The added recursive benchmark

and revisions received local checks without a second review. This AI assessment is not independent human peer review or certification of publishability.

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